D2CMFDet: Disparity-Guided Dynamic Cross-Modal Mamba Fusion for Multispectral Object Detection

Hai Yang, Xingdu Wu, Chenhai Wei, Yunfei Tong, Dan Zhou, Tianyi Zhou, Ting Xiao, Qian Zhang, Qin Zhou, Zhe Liu, and Zhe Wang, *Associate Member, IEEE*

***Abstract*—Existing multispectral object detectors struggle to balance wide-area cross-modal discrepancy modeling with high-resolution local detail preservation. To address this bottleneck, we propose D2CMFDet, a disparity-guided dynamic fusion framework that integrates learnable state-transition modeling with cross-modal feature calibration. Here, disparity denotes a channelwise feature-response discrepancy between RGB and infrared branches in the learned representation space, rather than geometric or stereo disparity, and it is not an estimate of modality reliability. At its core lies a unified Disparity-Guided Cross-Modal Mamba Fusion (DGCM) engine, which follows a cascaded enhance-then-fuse architecture. Within DGCM, the dynamic feature enhancement module (DFEM) embeds a diagonal local differential convolution (DLDConv), implemented as a structurally modulated convolution with a diagonal bias, to strengthen direction-sensitive local structural cues before global fusion. After this local enhancement stage, the calibrated modality features are converted into a constructed joint feature, which is processed by the Joint-Feature Mamba Fusion (JFMF) module through linear-complexity state-space modeling. This design explicitly separates two roles: DFEM preserves modality-specific local calibration, whereas JFMF performs efficient global modeling on the constructed joint feature. Extensive experiments on public RGBT datasets demonstrate that D2CMFDet achieves state-of-the-art or highly competitive performance in detection accuracy, inference speed, and robustness with favorable computational overhead. Code is available at https://github.com/jacksonwu09/D2CMFDet**

***Index Terms*—Cross-modal fusion, disparity-guided, dynamic feature enhancement, Multispectral object detection, state-space modeling.**

I. Introduction

OBJECT detection based on visible-light images has achieved remarkable success in applications like autonomous driving [1], urban security [2], and remote sensing [3], [4]. However, these methods remain highly sensitive to ambient illumination, frequently suffering severe performance degradation in low-light or adverse weather conditions. As illustrated in Fig. 1, infrared imaging effectively overcomes these limitations by capturing thermal radiation, making it irreplaceable for robust nighttime and complex-environment monitoring. Integrating these distinct sensors naturally introduces a critical physical challenge. In complex environments, visible-light observations may suffer from degraded target clarity, whereas the infrared branch can still retain distinguishable structural responses. This cross-modal inconsistency indicates that the two modalities provide complementary but sometimes inconsistent target-related responses for the same scene. In this work, modality disparity denotes the relative response discrepancy between RGB and infrared features in the learned representation space, rather than geometric or stereo disparity. This feature-space response-imbalance cue exposes where the two representations disagree and conditions shared-context allocation under detection supervision. Such disagreement marks the contested channels where naive equal fusion is more likely to inject conflicting context. By conditioning fusion on the current cross-modal response discrepancy, the framework performs adaptive shared-context allocation and disparity-conditioned feature calibration under challenging imaging conditions. This also explains why multispectral object detection is particularly valuable in challenging scenarios [5], [6].

Visible-infrared representation learning similarly exploits complementary cross-modality cues under changing appearance conditions [7].

Convolutional neural networks (CNNs) are effective at extracting local modality-specific representations in multispectral detection [8], [9]. However, in RGBT fusion, the central challenge is not merely local feature extraction, but how to support efficient cross-modal global interaction at relatively high resolutions while preserving the boundary-sensitive local structures required for object detection. Owing to their localized inductive biases (e.g., 3×3 kernels), CNN-based fusion models typically rely on progressively deeper stacking to enlarge the effective receptive field, which can limit the efficiency of long-range cross-modal interaction at shallow high-resolution stages. To address this limitation, Transformer-based methods have been introduced to perform

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Hai Yang, Xingdu Wu, Chenhai Wei, Yunfei Tong, Ting Xiao, Qian Zhang, Qin Zhou, Zhe Liu, and Zhe Wang are with the Department of Computer Science and Engineering, East China University of Science and Technology, Shanghai 200237, China (e-mail: wangzhe@ecust.edu.cn).

Dan Zhou is with The Second Affiliated Hospital of Zhejiang University School of Medicine, Zhejiang University, Hangzhou 310058, China.

Tianyi Zhou is with the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI 48109 USA.

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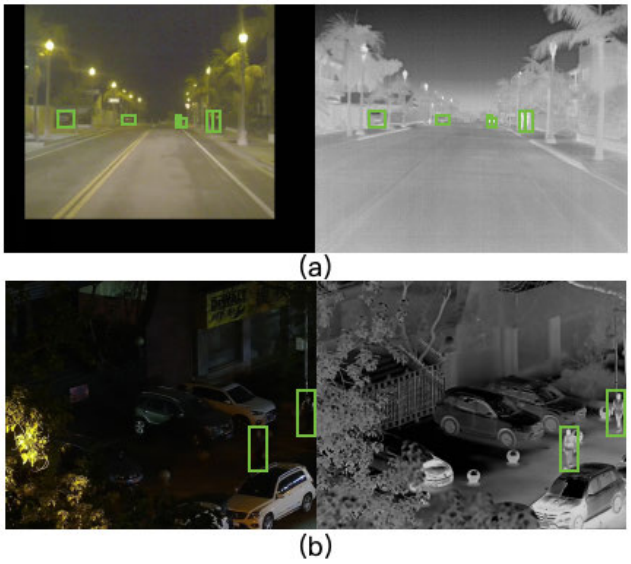


Fig. 1. Typical manifestations of modality disparity in nighttime (a) FLIR and (b) LLVIP scenes: RGB observations suffer from degraded target clarity, while infrared observations retain distinguishable structural responses, leading to relative response discrepancy between the two modalities.

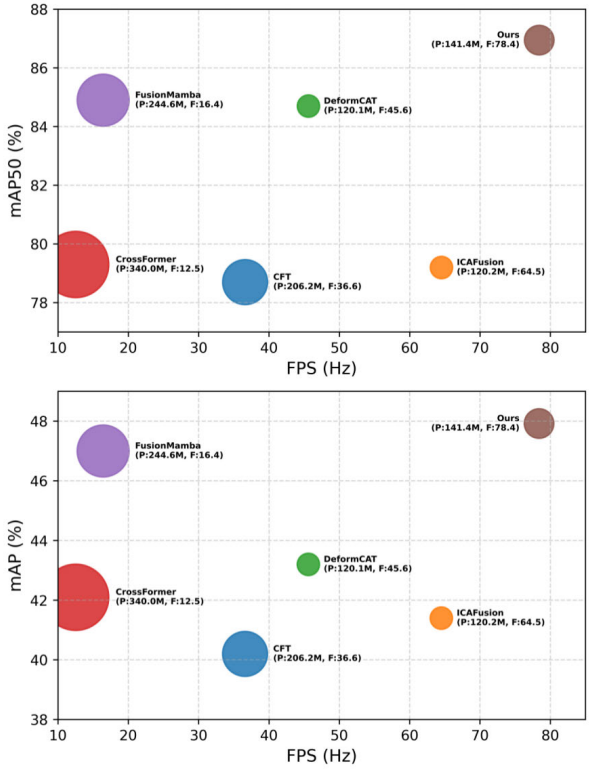


Fig. 2. Comparison of different multispectral object detection methods in terms of detection performance and inference speed (FPS). The bubble size represents the number of parameters for each method.

cross-modal interaction through attention mechanisms [10], [11], [12]. Compared with CNN-based fusion, they enable more direct global dependence modeling, but often introduce a different tradeoff under high-resolution dense prediction. In particular, directly applying self-attention to high-resolution feature maps incurs substantial computational cost because of its quadratic complexity ($O(N^2)$). As a result, practical architectures often adopt early downsampling or token compression to maintain tractable computation, which may in turn attenuate high-frequency structural details, particularly the thermal contours and fine textures that are crucial for infrared targets. Accordingly, the key challenge lies in developing a fusion architecture that can support efficient global dependence modeling without undermining the local structural fidelity required for multispectral object detection.

Mamba introduces selective state-space models (SSMs) for efficient long-range dependence modeling with linear computational complexity [13]. By leveraging a selective state evolution mechanism, Mamba provides strong global contextual representation capabilities that support semantic consistency in cross-modal fusion tasks. However, pure SSM-based visual modeling may still face challenges in preserving localized structures, especially when high-frequency contours, fine textures, and local boundary cues are critical for multispectral fusion [14], [15], [16]. This potential limitation is closely related to the inherent 1-D sequential scanning mechanism of visual state-space modeling: flattening 2-D spatial features into 1-D causal sequences may weaken the native local topological proximity of images, thereby reducing the direct structural constraint imposed by neighboring pixels. Consequently, compared with architectures that explicitly preserve 2-D spatial priors, pure SSM-based modeling can be more susceptible to feature smoothing when handling fine-grained local patterns. Therefore, how to compensate for the loss of local structural fidelity and suppress redundant cross-modal interference while preserving the linear-complexity advantage of SSM-based global dependence modeling remains a core challenge in multispectral object detection.

To address the above design challenge, we propose D2CMFDet, a disparity-guided cross-modal fusion framework that integrates learnable state-transition modeling with dynamic feature calibration. The proposed framework organizes local structural enhancement, disparity-guided dynamic gating, and joint-feature state-space fusion into a unified Disparity-Guided Cross-Modal Mamba Fusion (DGCM) engine. Within this hierarchy, diagonal local differential convolution (DLDConv) provides adaptive direction-sensitive local structural enhancement on early representations, establishing a stronger local basis for subsequent disparity-guided feature redistribution. Building upon this local compensation, the dynamic feature enhancement module (DFEM) uses cross-modal response discrepancy to regulate how the shared context is injected back into the two modality-conditioned streams. The calibrated modality features are then used to construct a joint driving feature for the Joint-Feature Mamba Fusion (JFMF) module, where efficient state-space scanning performs global modeling on the constructed joint feature at relatively high resolutions. Through the coordinated integration of these components, D2CMFDet establishes a unified dynamic fusion framework that jointly improves semantic consistency, cross-modal collaboration, and local structural preservation in multispectral object detection.

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Benefiting from the above design, D2CMFDet achieves strong robustness and favorable performance in multispectral detection. As illustrated in Fig. 2, the proposed framework provides a competitive balance between detection accuracy and inference efficiency. Relative to DeformCAT, D2CMFDet achieves gains of 4.7 percentage points in mAP and 2.2 percentage points in mAP50 while delivering a 71.9% increase in inference speed with only a 17.7% increase in parameter count. These results demonstrate the effectiveness of the proposed framework in balancing cross-modal detection accuracy and computational efficiency.

The main contributions of this paper are summarized as follows.

1) We propose D2CMFDet, a novel multispectral framework powered by a unified DGCM engine. DGCM follows a cascaded enhance-then-fuse architecture and is organized around two coordinated components, namely DFEM and JFMF, thereby balancing high-resolution local detail preservation with global spatial fusion efficiency.
2) Within DGCM, DFEM embeds DLDConv, a structurally modulated convolution with a diagonal bias, for initial local perception. It explicitly introduces directional high-frequency priors to strengthen target-related structural cues and enhance early local feature refinement.
3) As the global component of DGCM, JFMF performs linear-complexity global modeling on the constructed joint feature. It executes efficient joint-feature spatial fusion directly at high resolutions, complementing the aforementioned local structural compensation to ensure robust semantic consistency.
4) Extensive experiments on public RGBT datasets demonstrate that D2CMFDet achieves state-of-the-art or highly competitive performance. Under complex conditions, it exhibits strong robustness in multimodal target detection while maintaining favorable computational efficiency against existing approaches.

II. Related Work

A. Multispectral Object Detection

Multispectral object detection has recently achieved significant progress. Researchers have systematically explored various technical approaches to fully exploit the complementary characteristics of visible and infrared data. According to the stage at which information fusion occurs, current multispectral object detection methods can be broadly classified into three categories: pixel-level fusion [17], [18], [19], [20], feature-level fusion, and decision-level fusion [21], [22], [23].

Currently, research in multispectral object detection primarily focuses on feature-level fusion methods. This strategy typically involves independently extracting features from visible and thermal infrared images, followed by fusion at intermediate or high-level stages. In recent years, the introduction of attention mechanisms has significantly advanced the development of adaptive fusion strategies, enabling more effective extraction and integration of salient features from both modalities and thereby achieving deep fusion of multidimensional complementary information. For example, CFT [10] utilizes the self-attention mechanism of Transformers to achieve efficient intramodel and intermodal fusion; Zhang et al. [24] combine cross-modal and intramodal attention mechanisms to enhance discriminative feature representation; C2Former [25] integrates cross-modal attention with adaptive feature sampling to improve feature alignment and fusion accuracy; Wang et al. [26] design a two-branch backbone and a three-branch feature enhancement network to comprehensively extract shared and complementary information; GAFF [27] introduces guided attentive feature fusion to adaptively aggregate multispectral pedestrian features. These attention-based methods have advanced cross-modal representation learning, yet their effectiveness can still be limited in challenging RGBT scenarios characterized by severe modality discrepancy and high-resolution fusion demands. In particular, when self-attention is applied to dense prediction at relatively high resolutions, its substantial computational cost often necessitates early downsampling or token reduction, which may in turn weaken boundary-sensitive structural details. Moreover, some existing fusion strategies enhance robustness by modeling modality reliability under degraded conditions, typically through dedicated uncertainty estimation, confidence-aware branches, or explicitly designed reliability predictors. Our framework models bidirectional feature differences as a relative response-discrepancy cue learned under detection supervision and directly applies this cue to dynamic shared-context allocation during cross-modal interaction. This formulation emphasizes disparity-conditioned feature redistribution within the fusion process itself, together with local structural fidelity and adaptive cross-modal interaction.

B. Mamba

Mamba [13] is built on the structured state-space sequence model (S4) [28], optimizing the dynamics of hidden states to achieve efficient modeling and inference of long sequences, surpassing the mainstream Transformer architecture on natural language processing tasks. With its data-driven, dynamic mechanism and hardware-friendly design, the model significantly improves the processing efficiency of long-sequence data, demonstrating exceptional performance in natural language processing. In recent years, Mamba has been successfully introduced to computer vision tasks such as image classification, video understanding, and medical image segmentation, showcasing its powerful sequence modeling capabilities. For instance, VMamba [14] effectively captures spatial dependencies in 2-D images by introducing a cross-scanning algorithm, thereby achieving results superior to those of traditional Transformer architectures in tasks such as object detection, instance segmentation, and tracking; similarly, LocalMamba [15] further enhances local dependence capture by employing a windowed selective scan with dynamic direction search, effectively harmonizing fine-grained local details with global understanding. In the domain of multispectral object detection, the potential of Mamba has also been preliminarily explored; Li et al. [29] proposed a perturbation-adaptive diffusion and cross-modal Fusion-Mamba module to reconstruct degraded features under adverse

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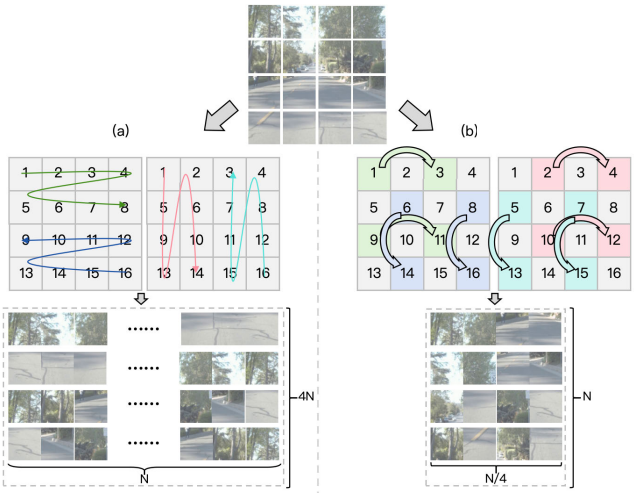


Fig. 3. Illustration of (a) SS2D and (b) ES2D mechanisms.

weather conditions; Fusion Mamba [30] enhanced the consistency of fused representations by reducing cross-modal feature discrepancies; AlignMamba [16] addressed the inherent limitations of sequential scanning by integrating local optimal transport and global distribution alignment mechanisms to learn robust cross-modal correspondences; JTMDet [31] further validated the advantages of Mamba in multispectral detection by employing a two-stage fusion strategy and hybrid module design. These works provide an important foundation for cross-modal fusion. Notably, recent visual Mamba studies such as LocalMamba and AlignMamba further suggest that pure state-space modeling alone may be insufficient for fine-grained local preservation, and therefore often introduce explicit local compensation or alignment mechanisms. Motivated by this observation, our framework extends Mamba-based cross-modal fusion by integrating disparity-guided dynamic redistribution with explicit local structural enhancement, so as to jointly support efficient global interaction and fine-grained local preservation.

CoMO [32] develops cross-Mamba interaction together with offset-guided fusion for multimodal object detection. Its design emphasizes interaction between modality-specific representations and spatial offset correction, whereas D2CMFDet performs disparity-conditioned shared-context redistribution before constructing a single joint feature for state-space modeling. This distinction further motivates the explicit separation between the DGCM engine-level fusion objective and the joint-feature scan topology used by JFMF.

III. Methodology

A. Preliminaries

Selective-scan mechanism: When applied to 2-D visual data, the standard Mamba architecture is constrained by its 1-D sequential modeling paradigm and thus struggles to capture complex spatial dependencies. VMamba introduces the selective scan 2-D (SS2D) mechanism, which unfolds image features into four traversal sequences along rowwise and columnwise forward and reverse routes, processes them with selective state-space modeling, and merges the restored spatial outputs. As illustrated in Fig. 3(a), this multidirectional formulation extends the effective receptive field while retaining linear sequence complexity.

EfficientVMamba [33] further optimizes this process via skip sampling, selectively scanning only sampled patches at fixed intervals, which substantially reduces computation. Compared to traditional SS2D, efficient 2-D scanning (ES2D) reduces the number of tokens per scan-and-merge operation from N to N/p^2, markedly improving feature extraction efficiency. As also shown in Fig. 3(b), ES2D achieves a balance between global context modeling and local detail preservation, thus ensuring both efficiency and comprehensive spatial representation in visual state-space modeling.

B. Architecture

Fig. 4 illustrates the overall architecture of the proposed D2CMFDet framework. Built upon the standard YOLOv5 architecture, this framework consists of a dual-branch backbone, a feature aggregation neck, and three detection heads. Both the RGB branch for processing visible images and the IR branch for handling thermal infrared images employ identical, pretrained CSPDarknet architectures as their backbones. For clarity, we denote the five levels of multiscale features extracted from the visible branch as C1–C5, and the corresponding five levels extracted from the infrared branch as P1–P5. To effectively combine complementary features and address modality discrepancy, we introduce a DGCM engine, denoted by the compact label DGCM. DGCM combines DFEM and JFMF within an engine-level fusion stage. Within this dynamic fusion framework, DFEM first performs local feature refinement through the coordinated action of an internal DLDConv-based local structural enhancement branch and a disparity-guided dynamic gating branch, after which the calibrated modality-conditioned features are forwarded to JFMF. JFMF constructs a joint driving feature from the calibrated RGB and thermal representations and applies state-space modeling to capture long-range dependencies on the constructed joint feature. Topologically, we adopt a halfway fusion strategy to hierarchically aggregate the deeper paired feature maps via DGCM. In this way, the proposed design strengthens global cross-modal interaction while preserving critical local structural information throughout the cascaded fusion process. Since the fused output feature maps are perfectly aligned in dimension with those generated by the single-stream CSPDarknet in vanilla YOLOv5, they can be seamlessly fed directly into the subsequent neck network for multiscale feature aggregation. Finally, three decoupled detection heads output the object categories and spatial bounding boxes, thereby achieving efficient end-to-end object detection.

C. Disparity-Guided Cross-Modal Mamba Fusion

As illustrated in Fig. 4, the proposed DGCM integrates multimodal features through a hierarchical “enhance-then-fuse” paradigm. Structurally, DGCM consists of two coordinated stages: DFEM for local modality-conditioned calibration and JFMF for subsequent global joint-feature modeling. Under this organization, DLDConv is embedded inside DFEM as the

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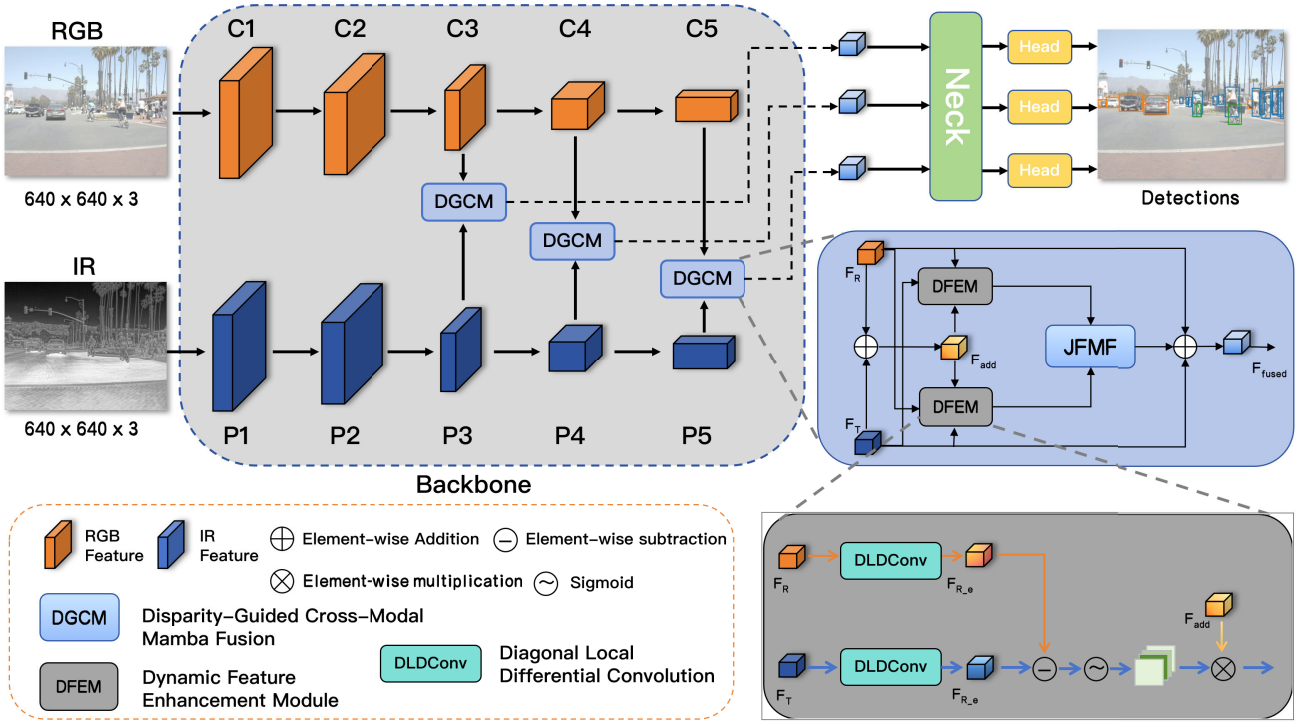


Fig. 4. Overall architecture of the proposed D2CMFDet framework. Built upon a YOLOv5 baseline, it employs a dual-branch backbone to extract hierarchical features from RGB and thermal images. In this paper, C1–C5 denote the five-stage features extracted by the visible branch, whereas P1–P5 denote the corresponding five-stage features extracted by the infrared branch. For cross-modal fusion, DGCM is inserted at the last three paired stages, namely (C3, P3), (C4, P4), and (C5, P5), whose channel dimensions are 256, 512, and 1024, respectively. DGCM is the engine-level fusion stage composed of DFEM and JFMF. DFEM is a composite local enhancement module consisting of a DLDConv-based local structural enhancement branch and a disparity-guided dynamic gating branch, while JFMF performs state-space modeling on the constructed joint feature. The zoomed-in view further clarifies this internal organization and shows DLDConv embedded within DFEM as an internal local enhancement operator before subsequent cross-modal interaction.

local structural enhancement operator, while JFMF performs selective state evolution on a constructed joint driving feature formed from the calibrated modality features. This decomposition establishes a clear functional transition from local structural refinement to global semantic integration.

Given input features $\mathbf{F_R}, \mathbf{F_T} \in \mathbb{R}^{\mathbf{C}\times\mathbf{H}\times\mathbf{W}}$ from the RGB and thermal branches, we first compute their elementwise addition $\mathbf{F}_{\text{add}}$ as a shared-context feature. Subsequently, the two modality-conditioned streams are jointly refined by DFEM, which internally combines a DLDConv-based local structural enhancement branch and a disparity-guided dynamic gating branch

$$(\mathbf{D}_1, \mathbf{D}_2) = \text{DFEM}\,(\mathbf{F_R}, \mathbf{F_T}, \mathbf{F}_{\text{add}}) \tag{1}$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ denote the calibrated modality-conditioned features produced by DFEM. These two features define the inputs from which JFMF constructs the joint driving feature. Finally, the fused feature $\mathbf{F}_{\text{fused}}$ is obtained by combining the JFMF output $\mathbf{Y}_{\text{JFMF}}$ with the initial aggregated feature via a residual connection

$$\mathbf{F}_{\text{fused}} = \mathbf{F}_{\text{add}} + \mathbf{Y}_{\text{JFMF}} \tag{2}$$

$$\mathbf{Y}_{\text{JFMF}} = \text{JFMF}\,(\mathbf{D}_1, \mathbf{D}_2) \tag{3}$$

where JFMF(·) denotes the proposed JFMF operation. This residual design helps preserve the original feature information while selectively integrating the enhanced global context, thereby improving representational robustness under severe modality degradation.

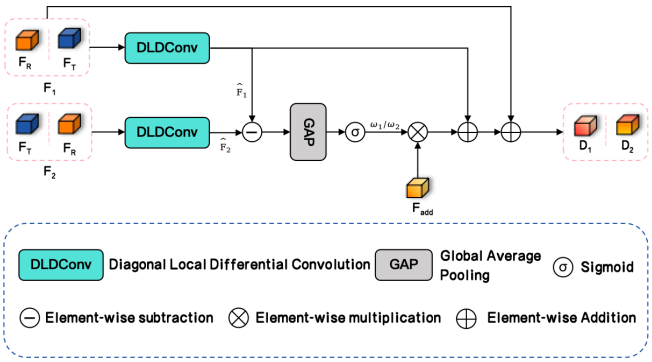


Fig. 5. Architecture of the proposed DFEM.

D. Dynamic Feature Enhancement Module

Existing cross-modal methods often struggle to simultaneously capture direction-sensitive local structures and perform adaptive shared-context modulation between modalities. To address this, we propose the DFEM, illustrated in Fig. 5. DFEM adopts a hierarchical nested design that couples local structural enhancement with disparity-guided dynamic modulation. At the foundational level, the embedded DLDConv is used to extract direction-sensitive local structural responses

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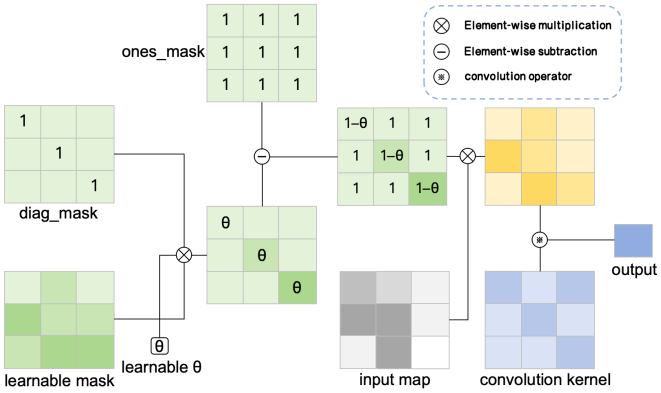


Fig. 6. Architecture of the proposed DLDConv, which is implemented as a structurally modulated convolution with a diagonal bias.

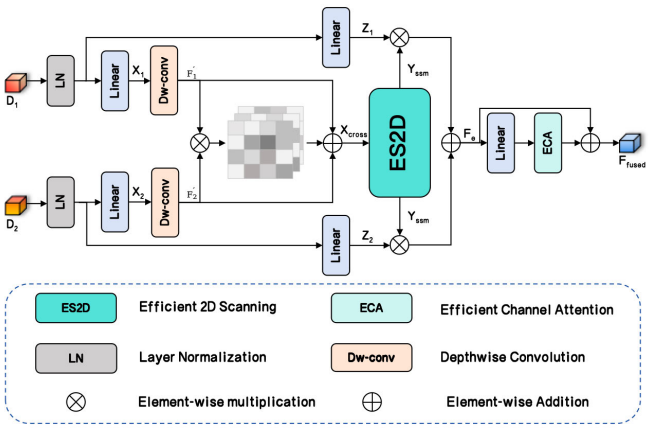


Fig. 7. Architecture of the proposed JFMF module.

with a diagonal bias, thereby strengthening contour and boundary perception in each modality stream. These enhanced local responses are subsequently compared between modalities through bidirectional feature differences to model intermodal local differential responses, which are then mapped into adaptive gating weights for dynamic shared-context modulation.

For compact notation within DFEM, let $\mathbf{F}_1 = \mathbf{F}_R$ and $\mathbf{F}_2 = \mathbf{F}_T$. We first extract direction-sensitive structural responses using the nested DLDConv

$$\hat{\mathbf{F}}_1 = \mathrm{DLDConv}(\mathbf{F}_1) \tag{4}$$

$$\hat{\mathbf{F}}_2 = \mathrm{DLDConv}(\mathbf{F}_2) \tag{5}$$

where $\hat{\mathbf{F}}_1$ and $\hat{\mathbf{F}}_2$ denote the structurally enhanced modality features.

DFEM further introduces a bidirectional difference modeling strategy to characterize cross-modal response discrepancy. The modality disparity in this work is quantified by the bidirectional feature differences $\hat{\mathbf{F}}_1 - \hat{\mathbf{F}}_2$ and $\hat{\mathbf{F}}_2 - \hat{\mathbf{F}}_1$. After channelwise global pooling, these signed differences provide a relative response-contrast cue in the learned feature space. The proposed gating weights are generated as

$$\mathbf{w}_1 = \sigma\left(\mathrm{GAP}\left(\hat{\mathbf{F}}_1 - \hat{\mathbf{F}}_2\right)\right) \tag{6}$$

$$\mathbf{w}_2 = \sigma\left(\mathrm{GAP}\left(\hat{\mathbf{F}}_2 - \hat{\mathbf{F}}_1\right)\right) \tag{7}$$

where GAP denotes global average pooling and σ is the Sigmoid function. By the linearity of GAP, the first gate can equivalently be written as $\mathbf{w}_1 = \sigma(\mathrm{GAP}(\hat{\mathbf{F}}_1) - \mathrm{GAP}(\hat{\mathbf{F}}_2))$. Thus, the cue is a signed channelwise pooled contrast. Because the two inputs are opposite in sign, the two gates satisfy a complementary relation in each channel, i.e., $\sigma(a) + \sigma(-a) = 1$. Therefore, $\mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^C$ serve as paired allocation gates derived from bidirectional cross-modal response contrasts.

The signed pooled contrast summarizes, for each sample and channel, the relative response ordering of the two structurally enhanced streams. This agreement-neutral construction yields equal allocation gates of 0.5 when the pooled responses are equal, and activates channelwise asymmetric allocation when signed response discrepancy emerges. DFEM maps this antisymmetric cue to adaptive shared-context redistribution. The two residual modality paths remain intact, while the complementary gates preserve a fixed shared-context budget: their two injections sum to the shared-context feature. GAP supplies one pooled value per image and channel, yielding a compact channel-level allocation signal; spatially resolved disparity modeling provides a natural future extension. A per-channel or illumination-conditioned sign and temperature for the allocation gate also provide natural extensions beyond the present parameter-free formulation. Compared with fixed 50/50 allocation, the learned allocation family retains equal allocation as a special case and permits channelwise asymmetric redistribution under detection supervision. The fixed-gate ablation and the gate-response visualization demonstrate the empirical benefit of learned allocation and show nonconstant, channel-dependent behavior.

Here, the joint context feature is defined as $\mathbf{F}_{\mathrm{add}} = \mathbf{F}_1 + \mathbf{F}_2$. The roles of $\mathbf{w}_1$ and $\mathbf{w}_2$ are to regulate how much shared context is injected into the two modality-conditioned streams. Under this formulation, $\mathbf{D}_1$ and $\mathbf{D}_2$ are separately calibrated modality-conditioned features formed on the visible and infrared branches, respectively. Accordingly, the proposed mechanism performs a soft, content-adaptive redistribution of shared context according to the signed pooled response contrast. The reallocated components are then anchored back to their respective spatial priors

$$\mathbf{D}_1 = \hat{\mathbf{F}}_1 + \mathbf{w}_1 \odot \mathbf{F}_{\mathrm{add}} + \mathbf{F}_1 \tag{8}$$

$$\mathbf{D}_2 = \hat{\mathbf{F}}_2 + \mathbf{w}_2 \odot \mathbf{F}_{\mathrm{add}} + \mathbf{F}_2 \tag{9}$$

where $\odot$ denotes elementwise multiplication. The two symmetric equations operate on distinct modality-conditioned streams and preserve separate refinement paths while assigning different amounts of shared context. Accordingly, $\mathbf{D}_1$ and $\mathbf{D}_2$ are two separately calibrated features that construct the joint driving feature for JFMF. The complementary constraint on $\mathbf{w}_1$ and $\mathbf{w}_2$ serves as a stream-level competitive redistribution rule before joint-feature state-space fusion.

E. Diagonal Local Differential Convolution

Conventional isotropic convolutions struggle to capture directionally sensitive structures. To address this limitation, we introduce DLDConv, illustrated in Fig. 6. Fundamentally, DLDConv is formulated as a structurally modulated

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Fig. 8. Qualitative detection results of four different scenes on the FLIR dataset. From top to bottom: (a) Ground truth. (b) CFT. (c) ICAFusion. (d) DeformCAT. (e) JTMDet. (f) D2CMFDet (ours). Orange bounding boxes indicate cars, blue boxes indicate persons, and green boxes indicate bicycles.

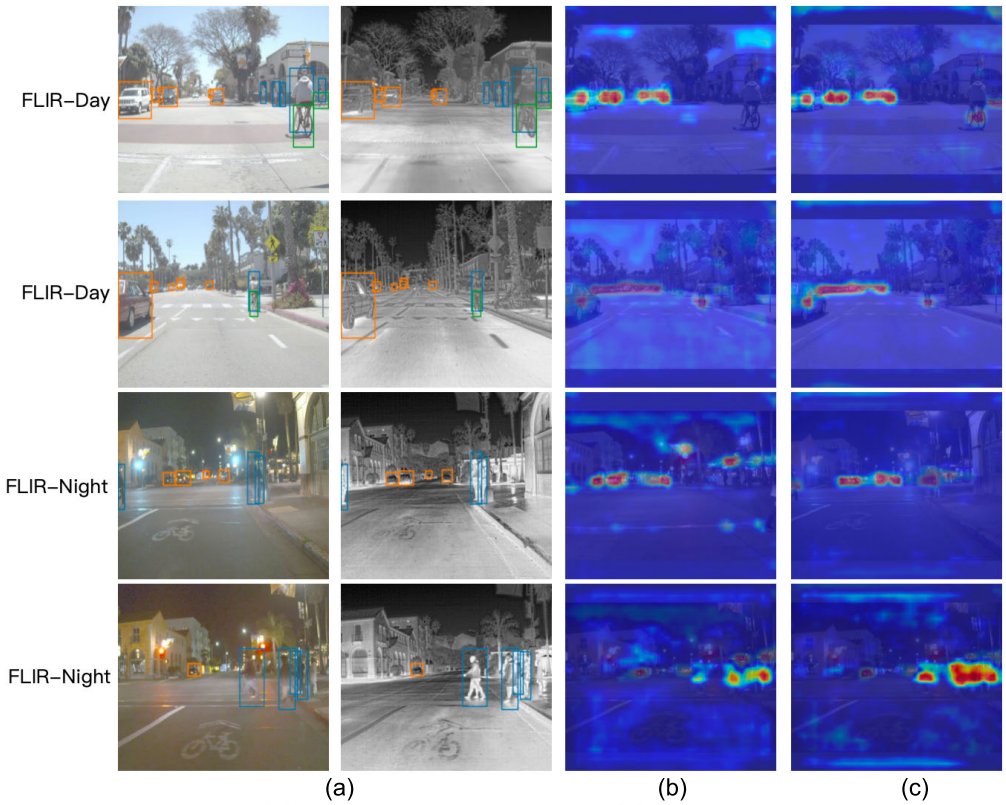


Fig. 9. Visualization results of attention maps on representative FLIR daytime and nighttime scenes. From left to right: ground truth in RGB and thermal images, heatmaps of the baseline method, and heatmaps of our proposed method. The rows correspond to representative FLIR scenes under different illumination conditions, and all comparisons are presented under consistent visualization settings to facilitate qualitative analysis of response distributions across different scene conditions. Orange bounding boxes indicate cars, blue boxes indicate persons, and green boxes indicate bicycles. (a) Ground truth. (b) Baseline. (c) Ours.

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT APPROACHES ON THE FLIR DATASET

Method	Year	Backbone	Data type	mAP50 (%)	mAP75 (%)	mAP (%)
FasterRCNN [40]	2015	ResNet50	RGB	64.9	21.1	28.9
FasterRCNN [40]	2015	ResNet50	IR	74.4	32.5	37.6
YOLOv5 [41]	2020	CSPDarknet53	RGB	67.8	25.9	37.8
YOLOv5 [41]	2020	CSPDarknet53	IR	73.9	35.7	39.5
GAFF [27]	2021	ResNet18	RGB+IR	72.7	30.9	37.3
CFT [10]	2021	CSPDarknet53	RGB+IR	78.7	35.5	40.2
CMAFF [42]	2022	CSPDarknet53	RGB+IR	76.6	–	39.8
UA-CMDet [5]	2022	ResNet-50	RGB+IR	78.6	–	–
CMPD [43]	2023	ResNet50	RGB+IR	69.4	–	–
CSSA [44]	2023	ResNet50	RGB+IR	79.2	37.4	41.3
MFPT [45]	2023	ResNet50	RGB+IR	80.0	–	–
MRD-YOLO [46]	2024	CSPDarknet53	RGB+IR	76.5	–	40.9
RSDet [47]	2024	ResNet50	RGB+IR	83.9	–	43.8
ICAFusion [11]	2024	CSPDarknet53	RGB+IR	79.2	36.9	41.4
CrossFormer [48]	2024	CSPDarknet53	RGB+IR	79.3	38.5	42.1
GM-DETR [49]	2024	HGNetv2	RGB+IR	83.9	42.6	45.8
EI^{2}Det [50]	2025	CSPDarknet53	RGB+IR	80.2	–	–
COFNet [51]	2025	CSPDarknet53	RGB+IR	83.6	41.7	44.6
DeformCAT [12]	2025	CSPDarknet53	RGB+IR	84.7	39.3	43.2
Fusion-Mamba [30]	2025	CSPDarknet53	RGB+IR	84.9	45.9	47.0
JTMDet [31]	2025	CSPDarknet53	RGB+IR	84.7	–	46.5
CIRDet [52]	2025	CSPDarknet53	RGB+IR	81.2	45.5	46.5
Fusion-DETR [53]	2025	ResNet50	RGB+IR	81.9	–	45.5
D2CMFDet(ours)	2026	CSPDarknet53	RGB+IR	**86.95**	**46.36**	**47.92**

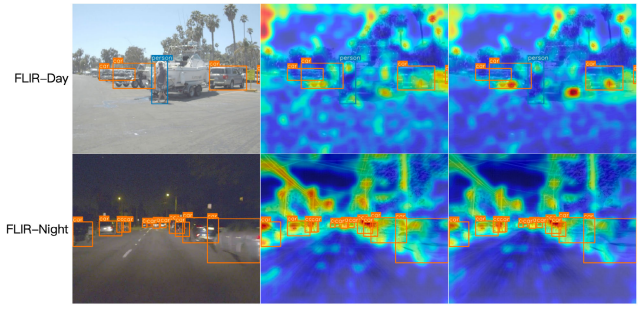


Fig. 10. Feature activation heatmaps extracted using Grad-CAM from the same output layer and visualized under the same normalization setting for fair comparison. The examples correspond to representative FLIR daytime and nighttime scenes. (Left) The original input image. (Middle) The feature responses of the baseline model, whose activations are more scattered and extend to irrelevant background regions such as the sky, road surfaces, and lamp reflections. (Right) The feature responses of D2CMFDet equipped with DFEM, where activations are more concentrated on the contours and key structural regions of valid targets.

TABLE II
QUANTITATIVE RESULTS FOR THE CVC-14 DATASET

Method	Year	Miss Rate (%)		
		All	Day	Night
AR-CNN [54]	2019	22.1	24.7	18.1
MBNet [36]	2020	21.1	24.7	13.5
MLPD [23]	2021	21.3	24.2	18.0
SCDNet [55]	2022	19.0	20.3	12.9
MCHE-CF [56]	2024	21.3	24.0	17.5
CPFM [57]	2024	15.5	20.6	**9.5**
DeformCAT [12]	2025	14.6	16.6	12.5
HFMIDet [58]	2025	13.4	16.2	11.7
D2CMFDet(ours)	2026	**13.17**	**12.17**	14.83

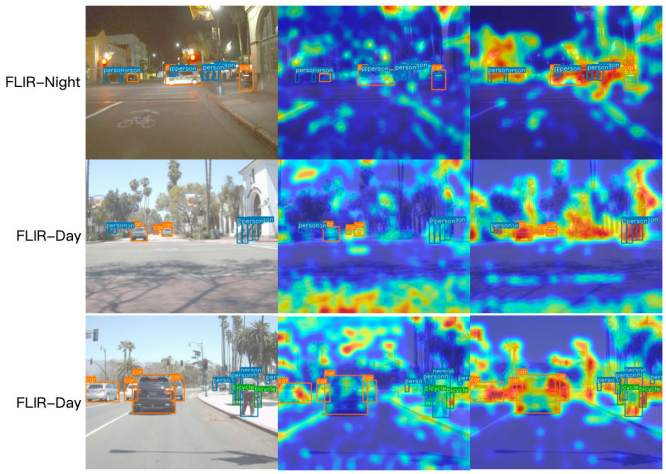


Fig. 11. Grad-CAM comparisons between the complete D2CMFDet without DLDConv and the full model under the same output layer and the same normalization setting. From left to right: original input image, feature responses of the complete D2CMFDet without DLDConv, and feature responses of the full model. Compared with the counterpart without DLDConv, the full model presents more concentrated activations around target contours and local structural regions.

convolution with a diagonal bias, while its core operation remains standard convolution, and the structured modulation is imposed on the convolution kernel prior to feature extraction. In the module name, "Diagonal" refers to the imposed diagonal structural bias on the convolution kernel, "Local" refers to the inherent local receptive-field property of convolution, and "Differential" does not denote a differential operator embedded in the convolution itself, but rather indicates the functional role of DLDConv in the subsequent DFEM pipeline, where the enhanced modality features are compared through bidirectional feature differences to model intermodal local differential responses. Accordingly, the module enhances

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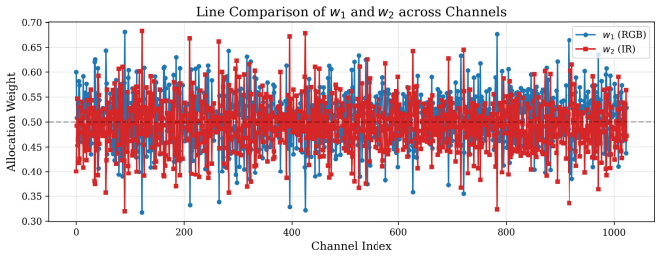


Fig. 12. Channelwise visualization of the learned dynamic gates $\mathbf{w}_1$ and $\mathbf{w}_2$ on the FLIR-3class validation set at the P5 fusion level. By construction, $\mathbf{w}_2 = 1 - \mathbf{w}_1$, so the two curves are exact mirror images about the equal-allocation reference value of 0.5; both are plotted to make the per-stream allocation directly readable.

TABLE III
QUANTITATIVE RESULTS FOR THE LLVIP DATASET

Method	Year	mAP50 (%)	mAP75 (%)	mAP (%)
GAFF [27]	2021	94.0	60.2	55.8
CFT [10]	2021	97.5	72.9	63.6
UA-CMDet [5]	2022	96.3	–	–
CSSA [44]	2023	94.3	66.6	59.2
M^{2}FNet [59]	2024	96.9	–	–
CrossFormer [48]	2024	97.4	75.4	65.1
RSDet [47]	2024	95.8	–	61.3
EI^{2}Det [50]	2025	98.0	73.2	63.9
Fusion-Mamba [30]	2025	97.0	–	64.3
COFNet [51]	2025	97.7	75.9	65.9
DeformCAT [12]	2025	**97.8**	77.1	66.1
COMO (YOLOv5s) [32]	2026	97.2	76.9	65.3
D2CMFDet(ours)	2026	97.71	**77.32**	**66.69**

TABLE IV
COMPARISON OF MODEL PARAMETERS AND INFERENCE SPEED

Method	Params (M)	FPS (Hz)
CFT	206.2	36.6
ICAFusion	120.21	64.5
DeformCAT	**120.13**	45.6
CrossFormer	340	12.5
Fusion-Mamba	244.6	16.4
JTMDet	237.3	28.9
D2CMFDet(ours)	141.4	**78.4**

direction-sensitive local structural responses and provides a stronger local structural basis for subsequent cross-modal differential modeling. This design improves the representation of high-frequency edges, fine textures, and boundary-sensitive cues that are important for cross-modal alignment.

More specifically, DLDConv preserves the standard convolutional accumulation rule and introduces a structured premodulation step on the convolution kernel. It utilizes a static diagonal mask $\mathbf{M}_{\text{diag}} \in \mathbb{R}^{k\times k}$ and a learnable channelwise modulation matrix $\mathbf{M}_{\text{learn}} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$ to selectively preprocess kernel weights before standard convolution is applied. This structured kernel modulation encourages the network to emphasize direction-sensitive local structures, thereby improving the modeling of fine-grained boundary and texture cues that are important for cross-modal alignment.

The structural convolution mask is formulated as

$$\mathbf{M}_{\text{mod}} = \mathbf{1}_{k\times k} - \theta \cdot \left(\mathbf{M}_{\text{learn}} \otimes \mathbf{M}_{\text{diag}}\right) \tag{10}$$

where $\mathbf{1}_{k\times k}$ is an all-ones baseline matrix, θ is a globally learnable modulation scalar, $\otimes$ denotes broadcast multiplication, and $\mathbf{M}_{\text{mod}}$ is the resulting directional mask.

Significantly, the proposed modulation is applied at the kernel level to the standard convolution weights $\mathbf{W}$ before feature extraction

$$\mathbf{W}_{\text{mod}} = \mathbf{W} \odot \mathbf{M}_{\text{mod}} \tag{11}$$

where $\odot$ denotes elementwise multiplication. This structural weight modulation dynamically imposes directional constraints during training, significantly enhancing diagonal pattern perception while keeping the inference-time convolutional computation unchanged.

Finally, the optimized kernel $\mathbf{W}_{\text{mod}}$ performs standard convolution on the input features $\mathbf{X}$, yielding the output $\mathbf{Y}$

$$\mathbf{Y} = \text{Conv2D}\left(\mathbf{X}, \mathbf{W}_{\text{mod}}\right). \tag{12}$$

This formulation preserves the core properties of standard convolution, such as translation invariance and local inductive biases. The output $\mathbf{Y}$ provides a stronger local foundation for subsequent cross-modal alignment.

F. Joint-Feature Mamba Fusion

In multimodal fusion tasks, a directly merged feature may fail to adequately represent the texture complementarity and structural asymmetry between RGB and infrared modalities. To address this limitation, we introduce a JFMF module, whose architectural design is illustrated in Fig. 7. The module is built upon the core state-space modeling formulation of Mamba. In particular, the calibrated modality features produced by DFEM are first transformed into modality-conditioned local representations, and then a constructed joint feature is formed as the input-dependent driving signal for selective state evolution. Therefore, the term "joint-feature" emphasizes that the selective scan is driven by this constructed joint feature. Under this formulation, the module performs joint-feature spatial modeling through multidirectional scanning, hidden-state recursion, output generation, and spatial restoration, while cross-modal interaction is encoded in the constructed driving feature and carried into the postscan composite gating.

Given the calibrated modality features $\mathbf{D}_1$ and $\mathbf{D}_2$ produced by DFEM, layer normalization is applied to each feature, followed by linear projection and channel splitting to generate the main branch features $\mathbf{X}$ and gating branch features $\mathbf{Z}$

$$\mathbf{X}_1, \mathbf{Z}_1 = \text{Split}\left(\text{Linear}\left(\text{LN}\left(\mathbf{D}_1\right)\right)\right) \tag{13}$$

$$\mathbf{X}_2, \mathbf{Z}_2 = \text{Split}\left(\text{Linear}\left(\text{LN}\left(\mathbf{D}_2\right)\right)\right) \tag{14}$$

where LN$(\cdot)$ denotes layer normalization, Linear$(\cdot)$ represents linear projection, and Split$(\cdot)$ denotes splitting the feature maps along the channel dimension.

Subsequently, depthwise separable convolution and nonlinear activation are employed on the main branch to extract local discriminative feature representations

$$\mathbf{F}_1' = \text{SiLU}\left(\text{DwConv}\left(\mathbf{X}_1\right)\right) \tag{15}$$

$$\mathbf{F}_2' = \text{SiLU}\left(\text{DwConv}\left(\mathbf{X}_2\right)\right) \tag{16}$$

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TABLE V

ABLATION STUDY ON THE EFFECTIVENESS OF THE MAJOR MODULES ACROSS THREE DATASETS. OWING TO THEIR DIFFERENT TOPOLOGICAL ROLES IN THE NETWORK, DIFFERENT ABLATION STRATEGIES ARE ADOPTED IN THIS BASIC ABLATION: DFEM IS EVALUATED BY DIRECT REMOVAL, AND JFMF IS EVALUATED WITH AN ELEMENTWISE ADDITION BASELINE

Method		FLIR-3class			CVC-14			LLVIP		
DFEM	JFMF	mAP50 (%)	mAP75 (%)	mAP (%)	MR-All (%)	MR-Day (%)	MR-Night (%)	mAP50 (%)	mAP75 (%)	mAP (%)
–	–	81.09	39.06	42.57	18.66	17.97	19.85	97.24	75.97	65.52
✓	–	86.24	43.83	46.59	15.14	16.43	**13.86**	97.37	76.49	65.71
–	✓	86.70	44.60	47.21	14.65	13.65	16.54	**97.88**	76.97	66.06
✓	✓	**86.95**	**46.36**	**47.92**	**13.17**	**12.17**	14.83	97.71	**77.32**	**66.69**

TABLE VI

CONTROLLED COMPARISON WITH CONFIDENCE- AND UNCERTAINTY-STYLE PROXY VARIANTS UNDER THE SAME FRAMEWORK. THE MODALITY CALIBRATION RULE IS CHANGED WHILE OTHER SETTINGS REMAIN UNCHANGED. PARAMS ARE REPORTED IN MILLIONS. FOR CVC-14, LOWER MR VALUES INDICATE BETTER PERFORMANCE. THE TWO PROXIES ARE LIGHTWEIGHT GATE-LEVEL CONSTRUCTIONS FOR THIS ABLATION, NOT REIMPLEMENTATIONS OF THE CITED FULL METHODS

	FLIR			
Method	mAP50	mAP75	mAP	Params (M)
Conf. proxy	86.75	45.69	47.45	141.4
Uncert. proxy	**87.22**	45.11	47.40	141.4
D2CMFDet	86.95	**46.36**	**47.92**	141.4
	CVC-14			
Method	MR-All	MR-Day	MR-Night	Params (M)
Conf. proxy	13.68	13.42	14.37	141.4
Uncert. proxy	13.96	14.46	**12.52**	141.4
D2CMFDet	**13.17**	**12.17**	14.83	141.4
	LLVIP			
Method	mAP50	mAP75	mAP	Params (M)
Conf. proxy	97.44	76.77	66.32	141.4
Uncert. proxy	97.47	76.03	66.02	141.4
D2CMFDet	**97.71**	**77.32**	**66.69**	141.4

TABLE VII

ABLATION STUDY ON DLDCONV MODULES

Method	*m*AP50 (%)	*m*AP75 (%)	*m*AP (%)
D2CMFDet(ours)	**86.95**	**46.36**	**47.92**
Removing DLDConv in RGB Branch	86.60	44.30	47.40
Removing DLDConv in IR Branch	86.20	44.60	47.40
Removing all DLDConv	85.90	44.70	46.90

TABLE VIII

ABLATION STUDY ON DIFFERENT STATE DIMENSIONS IN SSM

d_state	mAP50 (%)	mAP75 (%)	mAP (%)
4	86.21	43.19	46.58
8	86.29	45.71	47.57
12	**86.95**	**46.36**	**47.92**
16	86.28	45.00	47.32

TABLE IX

ABLATION STUDY ON THE POSITION OF DGCM IN D2CMFDET ON THE FLIR-3CLASS DATASET

(C3, P3)	(C4, P4)	(C5, P5)	mAP50 (%)	mAP75 (%)	mAP (%)
✓			84.10	41.93	44.70
	✓		82.85	39.43	43.11
		✓	82.18	38.86	42.95
✓	✓		86.51	43.72	46.32
✓		✓	83.27	41.20	44.26
	✓	✓	83.00	37.53	42.72
✓	✓	✓	86.95	46.36	47.92

TABLE X

COMPARATIVE EXPERIMENTS WITH SINGLE-STREAM ES2D ON THE FLIR, CVC-14, AND LLVIP DATASETS

Method	mAP50 (%)	mAP75 (%)	mAP (%)
D2CMFDet	**86.95**	46.36	**47.92**
Only ES2D	86.00	**46.40**	47.60
Method	Miss Rate (%)		
	MR–All	MR–Day	MR–Night
D2CMFDet	**13.17**	**12.17**	**14.83**
Only ES2D	14.81	13.94	15.25
Method	mAP50 (%)	mAP75 (%)	mAP (%)
D2CMFDet	**97.71**	**77.32**	**66.69**
Only ES2D	97.42	76.87	66.02

Note: Since ES2D is a single-stream architecture, the features from both modalities are combined via addition before being input into the ES2D module.

where DwConv(·) denotes depthwise separable convolution, SiLU(·) is the activation function, and $\mathbf{F}_1'$ and $\mathbf{F}_2'$ are the extracted local representations within JFMF.

To achieve deep feature alignment and correlation fusion, we construct a joint feature to provide the input-dependent driving signal for state-space modeling

$$\mathbf{X}_{\text{cross}} = \mathbf{F}_1' \otimes \mathbf{F}_2' + \mathbf{F}_1' + \mathbf{F}_2' \tag{17}$$

where $\otimes$ denotes elementwise multiplication and $\mathbf{X}_{\text{cross}}$ represents the constructed joint feature.

JFMF performs selective scanning on the single constructed joint feature. DFEM introduces cross-modal interaction before the scan through disparity-conditioned calibration; the multiplicative term $\mathbf{F}_1' \otimes \mathbf{F}_2'$ encodes intermodal correlation in the joint driving feature, and the two modality-conditioned projections $\mathbf{Z}_1$ and $\mathbf{Z}_2$ carry this information into postscan composite gating. Thus, DGCM integrates cross-modal calibration with the joint-feature scan topology of JFMF.

For joint-feature state-space modeling in the 2-D visual domain, $\mathbf{X}_{\text{cross}}$ is unfolded along four scanning directions to

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TABLE XI
EXPERIMENTAL COMPARISONS AGAINST DIAGONAL CONVOLUTION USING THE FLIR, CVC-14, AND LLVIP DATASETS

Method	mAP50 (%)	mAP75 (%)	mAP (%)
D2CMFDet	**86.95**	**46.36**	**47.92**
Diagonal Conv	86.60	44.20	46.70
Method	Miss Rate (%)		
	MR–All	MR–Day	MR–Night
D2CMFDet	**13.17**	**12.17**	**14.83**
Diagonal Conv	16.50	16.71	14.93
Method	mAP50 (%)	mAP75 (%)	mAP (%)
D2CMFDet	**97.71**	**77.32**	**66.69**
Diagonal Conv	97.23	75.41	65.58

Note: Diagonal Conv is implemented by retaining only the weights on the kernel's diagonal and setting all other positions to zero.

TABLE XII
COMPARISON BETWEEN DYNAMIC DISPARITY-GUIDED GATING AND A FIXED EQUAL-WEIGHT SETTING ON THE FLIR-3CLASS DATASET. HERE, Δ = FIXED − ADAPTIVE; NEGATIVE VALUES INDICATE LOWER PERFORMANCE FOR THE FIXED SETTING

Method	mAP50 (%)	mAP75 (%)	mAP (%)
Adaptive	**86.95**	**46.36**	**47.92**
Fixed (0.5/0.5)	86.81	44.87	47.42
Δ	-0.14	-1.49	-0.50

form continuous 1-D sequences for subsequent state evolution

$$\mathbf{x}_{\text{cross},d} = \text{Flatten}_d\left(\mathbf{X}_{\text{cross}}\right) \tag{18}$$

where $d \in \{1, 2, 3, 4\}$ denotes the scanning direction.

Following the standard selective scan formulation of SSMs, the continuous transition matrix $\mathbf{A}$ and input matrix $\mathbf{B}$ are discretized through the zero-order hold (ZOH) rule:

$$\bar{\mathbf{A}} = \exp\left(\Delta_t \mathbf{A}\right) \tag{19}$$

$$\bar{\mathbf{B}}_t = \left(\Delta_t \mathbf{A}\right)^{-1}\left(\exp\left(\Delta_t \mathbf{A}\right) - \mathbf{I}\right) \cdot \mathbf{B}_t \cdot \Delta_t \tag{20}$$

where Δ_t denotes the input-dependent step size and $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}_t$ are the corresponding discretized state-transition matrices. The hidden state is then recursively updated along each scan path as

$$\mathbf{h}_d^{(t)} = \bar{\mathbf{A}}\mathbf{h}_d^{(t-1)} + \bar{\mathbf{B}}_t \mathbf{x}_{\text{cross},d}^{(t)} \tag{21}$$

and the stepwise output is generated by

$$\mathbf{y}_d^{(t)} = \mathbf{C}_t \mathbf{h}_d^{(t)} \tag{22}$$

where $\mathbf{h}_d^{(t)}$ denotes the hidden state at step t, $\mathbf{C}_t$ is the parameterized output projection matrix, and $\mathbf{y}_d^{(t)}$ denotes the stepwise response of the SSM.

The 1-D responses are then unflattened back to the 2-D spatial domain and symmetrically aggregated to generate a shared joint-feature response $\mathbf{Y}_{\text{ssm}}$

$$\mathbf{Y}_{\text{ssm}} = \sum_{d=1}^{4} \text{Unflatten}_d\left(\mathbf{y}_d\right) \tag{23}$$

where $\text{Unflatten}_d(\cdot)$ restores the 1-D sequence to 2-D topological structures and $\mathbf{Y}_{\text{ssm}}$ is the shared spatial feature generated by the SSM scanning.

The shared response is then elementwise multiplied with the activated gating branches of each modality, and the resulting bimodal features are subsequently aggregated as follows:

$$\mathbf{F}_e = \mathbf{Y}_{\text{ssm}} \otimes \text{SiLU}\left(\mathbf{Z}_1\right) + \mathbf{Y}_{\text{ssm}} \otimes \text{SiLU}\left(\mathbf{Z}_2\right) \tag{24}$$

where $\mathbf{F}_e$ represents the aggregated interactive features.

Equation (24) applies a single composite gate, formed by the sum of the two activated modality-conditioned projections, to the shared SSM response. The postscan stage therefore carries modality-conditioned information into the shared response through composite gating.

Finally, the fused features undergo linear projection and adaptive channel attention through the ECA module, with residual connections further enhancing feature representation capability

$$\mathbf{Y}_{\text{JFMF}} = \text{ECA}\left(\text{Linear}\left(\mathbf{F}_e\right)\right) + \mathbf{F}_e \tag{25}$$

where $\text{ECA}(\cdot)$ denotes efficient channel attention and $\mathbf{Y}_{\text{JFMF}}$ represents the output of JFMF used in (2).

G. Loss Function

The loss function is consistent with that of the YOLO model, and the overall loss consists of three components: bounding box regression loss, classification loss, and confidence loss

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{box}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{obj}}. \tag{26}$$

Here, λ_1, λ_2, and λ_3 are the weights of the three loss terms, respectively. $\mathcal{L}_{\text{box}}$ adopts the generalized intersection over union (GIoU) loss, while $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{obj}}$ use cross-entropy loss and squared error loss, respectively.

IV. EXPERIMENTS

A. Dataset and Evaluation Metric

1) FLIR Dataset: The FLIR dataset is a challenging multispectral object detection benchmark encompassing both daytime and nighttime scenarios. Since the original version contains numerous misaligned RGB-thermal pairs, we adopt the latest aligned version [34], where unaligned pairs were manually removed. This refined dataset provides 5142 high-quality aligned pairs (4129 for training, 1013 for testing) across three categories: person, car, and bicycle.

2) CVC-14 Dataset: The CVC-14 dataset [35] provides diverse daytime and nighttime visible-infrared image pairs captured by a vehicle-mounted camera for pedestrian detection. Following the MBNet protocol [36], it is split into 1497 training pairs and 1304 testing pairs.

3) LLVIP Dataset: The LLVIP dataset [37] contains 15 488 strictly spatiotemporally aligned RGB-thermal image pairs specifically designed for pedestrian detection, ensuring highly reliable multimodal correspondence. It includes 12 024 training pairs and 3464 testing pairs.

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4) Evaluation Metric: For the FLIR and LLVIP datasets, we adopt three standard MS-COCO metrics: mAP, mAP@0.5, and mAP@0.75. As the primary and most stringent criterion, mAP comprehensively aggregates the mean average precision across multiple IoU thresholds, where a higher value indicates superior detection performance. For the CVC-14 dataset, we employ the log-average miss rate (MR) [38]. The MR is computed by averaging the MRs across the false positives per image (FPPI) range from 10^{-2} to 10^{0}, with a lower value denoting better performance. Following standard protocols, the evaluation encompasses three distinct settings: "All," "Day," and "Night."

B. Implementation Details

The proposed D2CMFDet framework is implemented in PyTorch [39], and all models are trained and evaluated on a single NVIDIA RTX 4090 GPU. The visible and infrared branches are instantiated as two parallel single-stream YOLOv5l backbones, both initialized with official MS-COCO pretrained weights. Since the infrared input is originally single-channel, it is replicated to three channels before being fed into the network so as to match the input format of the pretrained YOLOv5l weights. Unless otherwise specified, all input images are resized to 640×640, and the batch size is set to 8. We adopt SGD as the optimizer with a weight decay of 5×10^{-4}, and use a warm-up plus cosine annealing learning-rate schedule. The initial learning rate is set to 0.001 for FLIR and 0.01 for both CVC-14 and LLVIP. The number of training epochs is set to 15, 30, and 30 for FLIR, CVC-14, and LLVIP, respectively. In JFMF, the projection dimension d_{model} is matched to the channel dimension of the attached fusion stage, the state dimension d_{state} is fixed to 12, and the internal expand ratio is set to 2. The overall loss follows the standard YOLO detection objective, which consists of bounding box regression loss, classification loss, and confidence loss. To improve generalization, we further apply random scaling, spatial translation, horizontal flipping, color jittering, and mosaic augmentation during training.

Unless explicitly identified as as-published results, quantitative results for D2CMFDet and its in-framework variants are reported from a single representative training run under the stated protocol. No multiseed averaging or statistical significance testing is performed; accordingly, subpoint differences are interpreted descriptively.

C. Comparisons

To comprehensively evaluate the proposed framework's effectiveness, we conducted systematic comparative experiments against various mainstream state-of-the-art methods on three large-scale public datasets: FLIR, CVC-14, and LLVIP.

Quantitative Results for FLIR: As shown in Table I, our method achieves strong performance on the FLIR dataset, attaining 86.95%, 46.36%, and 47.92% in terms of mAP50, mAP75, and mAP, respectively. Compared with the recent competitive methods DeformCAT and Fusion Mamba, our approach yields consistent improvements across all three metrics. In particular, it improves mAP by 4.72 percentage points over DeformCAT and by 0.92 percentage points over Fusion Mamba while also maintaining the best mAP50 and mAP75 performance. These results further validate the effectiveness and robustness of the proposed framework for multispectral object detection.

Quantitative Results for CVC-14: As shown in Table II, our method achieves MRs of 13.17%, 12.17%, and 14.83% under the All, Day, and Night settings, respectively. In both the All and Day scenarios, D2CMFDet attains the lowest MR among the compared methods; relative to DeformCAT, the reductions are 1.43 and 4.43 percentage points. On the Night subset, its MR of 14.83% is higher than those of CPFM (9.5%), HFMIDet (11.7%), DeformCAT (12.5%), SCDNet (12.9%), and MBNet (13.5%). The CVC-14 results are therefore interpreted as a rule-dependent Day–Night allocation tradeoff on this weakly aligned benchmark. In the same-framework comparison, the signed-contrast rule attains the lowest MR-All and MR-Day (13.17% and 12.17%), whereas the inverse-response proxy attains a lower MR-Night (12.52%) at the cost of a higher MR-Day (14.46%). Low-light evidence is accordingly anchored to the strictly aligned all-low-light LLVIP benchmark and the FLIR-Night visual analysis. Night-adaptive and spatially resolved disparity cues are promising directions for future work.

Quantitative Results for LLVIP: As shown in Table III, our method achieves competitive performance on the LLVIP dataset, attaining 97.71%, 77.32%, and 66.69% in terms of mAP50, mAP75, and mAP, respectively. Compared with the recent state-of-the-art method DeformCAT, our approach yields improvements of 0.22 percentage points in mAP75 and 0.59 percentage points in mAP while maintaining highly competitive mAP50 performance. These results highlight competitive detection quality, especially on the stricter mAP75 and mAP metrics, and further support the effectiveness of the proposed framework for multisource information fusion under challenging illumination and fine-grained detection conditions.

In addition to the generic comparison with mainstream multispectral detectors, Tables I–III also include several published uncertainty-, confidence-, or proposal-guided methods, including UA-CMDet, CMPD, and CIRDet on FLIR, MCHE-CF and CPFM on CVC-14, and UA-CMDet on LLVIP. CPFM is categorized as a proposal-guided baseline. The tables also include JTMDet on FLIR and COMO on LLVIP as published Mamba-based context. On FLIR, D2CMFDet obtains 86.95/46.36/47.92 in mAP50/mAP75/mAP. CIRDet, the strongest listed uncertainty- or confidence-related detector with all three reported FLIR metrics, obtains 81.2/45.5/46.5, corresponding to margins of 5.75, 0.86, and 1.42 percentage points, respectively. UA-CMDet and CMPD report only mAP50 values of 78.6% and 69.4%. JTMDet reports 84.7 mAP50 and 46.5 mAP on FLIR, with mAP75 unavailable in its reported results. COMO (YOLOv5s) reports 97.2/76.9/65.3 on LLVIP under its published 12 025/3463 train/test split, whereas the protocol used here is 12 024/3464. Together, these as-published entries provide broader context for recent Mamba-based fusion designs. Because the reported values use different backbones and training protocols, Section IV-E complements them with the same-framework controlled

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comparison that holds the backbone, fusion locations, datasets, training protocol, detection head, and loss function fixed while changing the modality calibration cue.

Comparison of Model Parameters and Inference Speed with Mainstream Methods: As shown in Table IV, D2CMFDet maintains 141.4M parameters and achieves a high inference speed of 78.4 FPS, showing a favorable balance between model complexity and efficiency. The parameter and FPS comparison is based on the same RTX 4090 evaluation environment and the common 640×640 input setting described above unless otherwise specified. For FPS measurement, we use a per-image inference protocol with batch size 1; the timed region covers model forward propagation and NMS postprocessing, while data loading, visualization, and result saving are excluded. This demonstrates that the proposed dynamic feature enhancement and efficient state-space modeling mechanisms improve the model's practical inference performance.

D. Visual Results

As shown in Fig. 8, existing mainstream methods (including CFT, ICAFusion, DeformCAT, and JTMDet) still suffer from missed detections and false positives in challenging scenes. The proposed D2CMFDet shows stronger robustness and detection accuracy, effectively identifying targets missed by other methods while reducing false positives.

Fig. 9 presents representative attention-map visualizations on the FLIR dataset, illustrating the proposed method's capability to focus on discriminative regions under both daytime and nighttime conditions. In the FLIR-Day scenes, the comparison highlights how different methods allocate responses over road surfaces, sky regions, and surrounding context when multiple targets coexist. In the FLIR-Night scenes, the visualization further reflects the effect of limited visible-light quality and complex illumination conditions on cross-modal perception. The baseline model tends to produce relatively strong responses in nontarget regions, which weakens the concentration on semantically relevant structures. On the contrary, D2CMFDet yields more concentrated responses on target-related regions across representative daytime and nighttime examples, indicating improved discrimination between salient objects and irrelevant background activations. Nevertheless, in highly cluttered scenes or under weak target-background contrast, some background responses may still remain, suggesting that DFEM improves redundant-context suppression but does not eliminate all distracting activations. These visual results provide complementary evidence for the robustness of the proposed fusion design under challenging illumination and cluttered-scene conditions.

To provide an interpretable and reproducible analysis of the internal mechanisms, we extract feature activation heatmaps using the Grad-CAM visualization method from the same output layer of the corresponding modules and compare them under the same normalization setting, as shown in Fig. 10. These visual comparisons are intended to reflect the effect of DFEM on cross-modal feature responses. In the baseline model (middle column), the activation distribution is relatively scattered and extends to nontarget regions such as the sky and road surfaces in daytime scenes, while in nighttime scenes it also responds to visually distracting background regions under complex illumination. By contrast, the model equipped with DFEM (right column) exhibits more concentrated activations around the contours and key structural regions of pedestrians and vehicles. This qualitative difference suggests that DFEM improves the discrimination of target-related responses and reduces the tendency of feature activations to spread over irrelevant background areas. This visual trend is also consistent with the quantitative gains reported in Table V: on the FLIR-3class dataset, introducing DFEM improves mAP from 42.57% to 46.59% when JFMF is removed, and from 47.21% to 47.92% when JFMF is enabled, indicating that the more target-focused responses produced by DFEM are accompanied by measurable detection improvements.

To further provide representative qualitative evidence for the effect of local structural compensation, we additionally compare the complete D2CMFDet with its counterpart obtained by removing all DLDConv modules. The Grad-CAM visualizations are extracted from the same output layer and compared under the same normalization setting, as shown in Fig. 11. The first column presents the original input images, the second column shows the responses of the complete D2CMFDet without DLDConv, and the third column shows the responses of the full model. Compared with the counterpart without DLDConv, the full model exhibits more focused activations on pedestrians, vehicles, and their key structural regions, with stronger concentration around target contours and local boundaries. This qualitative phenomenon is consistent with the quantitative gains obtained in the DLDConv ablation, indicating that explicit local structural compensation helps subsequent joint-feature state-space interaction preserve fine-grained local structural sensitivity.

E. Ablation Study

1) Effectiveness of the Proposed Modules: We verify the effectiveness of the two proposed core components through systematic ablation experiments in Table V. To rigorously evaluate their individual contributions, we employ two distinct ablation strategies: direct removal for DFEM and baseline-equivalent replacement using elementwise addition for JFMF.

On the FLIR dataset, independently introducing either DFEM or JFMF improves detection performance. In particular, JFMF makes a notable contribution to mAP75, increasing it from 39.06% to 44.60%, which demonstrates its effectiveness for high-quality localization through efficient global state-space interaction. When both components are introduced together, mAP50, mAP75, and mAP reach 86.95%, 46.36%, and 47.92%, respectively. This result indicates a clear synergistic effect between fine-grained structural modeling and joint-feature global fusion. On the CVC-14 dataset, the full model jointly reduces MR-All from 18.66% to 13.17% and MR-Day from 17.97% to 12.17% relative to the baseline. The DFEM-only setting attains the lowest MR-Night of 13.86%; adding JFMF therefore trades part of the Night-subset miss-rate reduction for stronger All and Day performance. This subset-specific tradeoff is reported explicitly and is consistent with the rule-dependent Day–Night behavior discussed above. On the LLVIP dataset, both components also contribute

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steadily to the overall performance improvement. JFMF raises mAP50%–97.88%, highlighting its value in preserving critical details and maintaining robust feature alignment during cross-modal fusion. Although the JFMF-only setting obtains a slightly higher mAP50 than the full model under near-saturated mAP50 conditions, the full model achieves better mAP75 and mAP, suggesting stronger high-quality localization and overall detection quality. When both components are jointly employed, mAP75 reaches 77.32%, and mAP increases to 66.69%, which further verifies that the proposed method improves both semantic consistency and local detail perception in complex environments.

2) Controlled Comparison With Response-Derived Calibration Proxies: To isolate the calibration cue, we replace only DFEM's gate rule with two parameter-free proxies inspired by confidence-based and uncertainty-aware fusion [5], [22], [43], [56]. The confidence proxy normalizes pooled single-modality responses

$$\mathbf{s}_1 = \sigma\left(\mathrm{GAP}\left(\mathbf{x}_1\right)\right) \tag{27}$$

$$\mathbf{s}_2 = \sigma\left(\mathrm{GAP}\left(\mathbf{x}_2\right)\right) \tag{28}$$

$$\left[\mathbf{w}_1, \mathbf{w}_2\right] = \mathrm{Softmax}_{\mathrm{mod}}\left(\left[\mathbf{s}_1, \mathbf{s}_2\right]\right). \tag{29}$$

The uncertainty proxy assigns larger weights to smaller nonnegative response scores

$$\mathbf{u}_1 = \mathrm{Softplus}\left(\mathrm{GAP}\left(\mathbf{x}_1\right)\right) \tag{30}$$

$$\mathbf{u}_2 = \mathrm{Softplus}\left(\mathrm{GAP}\left(\mathbf{x}_2\right)\right) \tag{31}$$

$$\mathbf{w}_1 = \frac{\exp\left(-\mathbf{u}_1\right)}{\exp\left(-\mathbf{u}_1\right) + \exp\left(-\mathbf{u}_2\right)} \tag{32}$$

$$\mathbf{w}_2 = \frac{\exp\left(-\mathbf{u}_2\right)}{\exp\left(-\mathbf{u}_1\right) + \exp\left(-\mathbf{u}_2\right)}. \tag{33}$$

All other components and training settings remain unchanged, so the three variants have the same parameter count. Because both proxies rely on global pooling and parameter-free normalization, the comparison isolates the response-to-weight mapping without changing model capacity.

As shown in Table VI, under the matched 141.4M-parameter setting, D2CMFDet achieves the best FLIR mAP75 and mAP, the best CVC-14 MR-All and MR-Day, and the best three LLVIP metrics. The uncertainty proxy performs best on FLIR mAP50 and CVC-14 MR-Night. D2CMFDet uses an anti-symmetric signed contrast, whereas the proxies transform the pooled branch responses independently before normalization. Overall, the signed pooled contrast is competitive across datasets without adding a prediction branch.

In a separate FLIR experiment, a learnable uncertainty head adds 1.38M parameters and obtains 86.44%, 43.90%, and 46.87% mAP50, mAP75, and mAP. The parameter-free disparity gate obtains 86.95%, 46.36%, and 47.92%, respectively. This experiment is reported separately because the model capacities differ.

3) Effect of the DLDConv Module: To further isolate the contribution of the local structural operator inside DFEM, we conduct direct-removal ablations on DLDConv, as summarized in Table VII. In particular, DLDConv is removed from the RGB branch, the IR branch, or both branches, while the remaining network structure is kept unchanged. Since DLDConv functions as an embedded local enhancement operator rather than a mandatory fusion interface, direct removal constitutes the most appropriate basic ablation setting for this module. Experimental results show that all removal settings lead to consistent performance degradation, while the complete model achieves the best results across all metrics. Removing DLDConv from either modality weakens the model's ability to preserve direction-sensitive boundaries and fine-grained structural cues, and removing all DLDConv modules causes the most evident overall degradation. Notably, when all DLDConv modules are removed, the mAP on FLIR-3class drops from 47.92% to 46.90%. This performance gap provides direct quantitative evidence that subsequent joint-feature state-space modeling alone is insufficient to fully preserve fine-grained local structures in the absence of explicit local compensation. On the contrary, the local structural enhancement introduced by DLDConv helps recover boundary-sensitive and texture-aware representations before global cross-modal fusion. Therefore, this ablation serves as the primary quantitative support for the necessity of local structural compensation in our joint-feature state-space fusion framework.

4) Effect of the SSM State Dimension: To evaluate the impact of the state dimension in the SSM on detection performance, we conducted an ablation study as shown in Table VIII. With all other modules unchanged, the state dimension d_state was set to 4, 8, 12, and 16 to assess its influence on detection accuracy. The experimental results show that when d_state is 4, the model achieves 86.21% mAP50, 43.19% mAP75, and 46.58% mAP. Increasing the state dimension to eight improves all metrics, with mAP rising to 47.57%, indicating that a higher dimensional state representation helps capture richer spatiotemporal features. Furthermore, increasing d_state to 12 yields the best performance, reaching 86.95%, 46.36%, and 47.92% on mAP50, mAP75, and mAP, respectively. However, when the dimension is increased to 16, overall performance slightly declines, suggesting that an excessively high state dimension may introduce redundancy and affect the model's generalization. Therefore, we adopt d_state = 12 as the default setting to balance accuracy and computational efficiency.

5) Effect of DGCM Position on FLIR-Aligned Dataset: As shown in Table IX, the baseline model achieves an mAP of 42.57%. When DGCM is applied to different paired stages individually, placing it at (C3, P3) yields the greatest improvement among single-stage configurations, increasing the mAP to 44.70%. Among dual-stage configurations, the {(C3, P3), (C4, P4)} combination achieves the best performance with an mAP of 46.32%, clearly higher than the other two-stage combinations. Notably, the triple-stage configuration {(C3, P3), (C4, P4), (C5, P5)} attains the highest performance across all metrics, with mAP50 reaching 86.95%, mAP75 reaching 46.36%, and mAP reaching 47.92%. This observation can be attributed to the complementary roles of different paired stages: the high-resolution paired stage (C3, P3) preserves precise spatial boundaries, and through the cross-stage cascade, these high-fidelity localization priors are progressively propagated to deeper fusion stages, explicitly compensating for the boundary blurring and semantic drift that tend to

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accumulate toward (C5, P5). Consequently, deploying DGCM across all three paired stages {(C3, P3), (C4, P4), (C5, P5)} achieves the best balance between performance and computational complexity, and this configuration is adopted as the default in subsequent experiments.

6) Effectiveness of the Joint-Feature State-Space Fusion Design: To characterize the contribution of the JFMF design, we compare it with a single-stream ES2D baseline. As shown in Table X, the ES2D baseline first merges the two modalities by simple addition and then performs scanning with a single hidden-state evolution path. D2CMFDet reports higher primary aggregate metrics across the three datasets while preserving efficient global modeling. In particular, compared with the ES2D baseline, the proposed model reports mAP differences of 0.32 percentage points on FLIR and 0.67 percentage points on LLVIP, together with a 1.64-percentage-point lower MR-All on CVC-14. For completeness, the ES2D baseline is marginally higher on FLIR mAP75 (46.40 versus 46.36, a 0.04-point difference); under the single-run protocol disclosed in Section IV-B, we treat this result as a tie rather than as a meaningful difference. These descriptive single-run results are consistent with a contribution from joint-feature construction beyond simple addition under the same overall setting. The results attribute the observed pattern to the combination of DFEM-based modality-conditioned calibration and JFMF-based joint-feature state-space fusion for cross-modal collaboration and long-range alignment.

7) Experimental Comparison Between DLDConv and Traditional Diagonal Convolution: To explicitly verify that the gain of DLDConv does not merely come from imposing a diagonal prior, we further replace DLDConv with traditional static diagonal convolution for innovation-boundary validation. As shown in Table XI, we compare the detection performance of DLDConv with traditional diagonal convolution across three major datasets. The traditional diagonal convolution introduces only a fixed directional prior and lacks adaptive modulation capability, which limits its ability to model complex local structures under varying multispectral conditions. On the contrary, the proposed DLDConv adopts learnable kernel modulation and is embedded into the disparity-guided enhancement pipeline of DFEM. Quantitatively, compared with traditional diagonal convolution, DLDConv improves mAP by 1.22 percentage points on FLIR and 1.11 percentage points on LLVIP while reducing MR-All by 3.33 percentage points on CVC-14. These results show that the gain of DLDConv does not come merely from a static diagonal prior, but from its dynamic structural enhancement mechanism for cross-modal local representation learning.

8) Effect of Dynamic Disparity-Guided Gating: To further examine the role of the complementary gates $\mathbf{w}_1$ and $\mathbf{w}_2$, we compare the learned dynamic allocation with a fixed equal-weight setting, i.e., $\mathbf{w}_1 = \mathbf{w}_2 = 0.5$ for all samples and channels. This fixed setting satisfies the complementary constraint and replaces sample-adaptive redistribution with static shared-context allocation to the two modality branches before entering JFMF.

As reported in Table XII, this representative validation on the FLIR-3class dataset shows that the learned dynamic gates achieve stronger performance than fixed equal weights. The complete model obtains 86.95/46.36/47.92 in mAP50/mAP75/mAP, while the fixed-weight variant obtains 86.81/44.87/47.42. In particular, the 1.49 percentage-point gain in mAP75 indicates that dynamic stream-level redistribution before joint-feature state-space fusion is especially important for preserving fine-grained localization quality. Both settings satisfy the same complementary constraint, and this comparison attributes the performance gain to disparity-guided dynamic modulation of two separate modality-conditioned inputs prior to subsequent joint-feature state-space interaction.

To further analyze the behavior of the dynamic gates, we visualize the channelwise responses of $\mathbf{w}_1$ and $\mathbf{w}_2$ on the FLIR-3class validation set at the P5 fusion level. As shown in Fig. 12, the two gates exhibit evident channel-dependent variations with nonconstant response patterns. The gates deviate only moderately from 0.5, approximately within the range [0.3, 0.7], yet this moderate channelwise asymmetry already accounts for the 1.49-point mAP75 gain over fixed allocation in Table XII. This observation indicates differentiated channelwise gating and adaptive allocation of shared context to the two modality streams. Combined with the fixed-weight comparison in Table XII and the controlled comparison with confidence- and uncertainty-style proxy variants in Table VI, this representative visualization further shows that the effectiveness of the proposed mechanism lies in response-adaptive shared-context modulation induced by signed pooled cross-modal contrast.

V. Conclusion

In this paper, we propose D2CMFDet, a disparity-guided dynamic fusion framework for multispectral object detection. The proposed method is designed to improve the balance between efficient joint-feature global modeling and fine-grained local structural preservation. In particular, DLDConv, embedded within DFEM as a structurally modulated convolution with a diagonal bias, enhances direction-sensitive local structural cues and provides a stronger local basis for subsequent cross-modal interaction. Meanwhile, DFEM performs disparity-guided stream-level redistribution of shared context, enabling the two modality-conditioned branches to be adaptively calibrated before joint-feature state-space fusion. Built upon linear-complexity state-space modeling, JFMF further supports efficient long-range global modeling at relatively high resolutions by performing selective scanning on the constructed joint feature. Extensive experiments on FLIR, CVC-14, and LLVIP demonstrate that the proposed framework achieves state-of-the-art or highly competitive performance under diverse multispectral detection settings while maintaining favorable inference efficiency. The controlled proxy comparison shows that, at matched capacity, signed channelwise pooled response contrast is a competitive parameter-free allocation rule. Together with the in-framework learnable uncertainty-head check, this evidence supports the limited conclusion that a dedicated uncertainty-prediction branch is not necessary to obtain this level of calibration in the present framework, rather than a universal superiority claim over uncertainty-aware alternatives. Separately, the

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published benchmark comparisons indicate that D2CMFDet retains competitive detection quality without such a dedicated branch. These results suggest that the coordinated integration of local structural enhancement, disparity-guided dynamic modulation, and joint-feature state-space fusion provides an effective framework for robust multispectral object detection in challenging environments.

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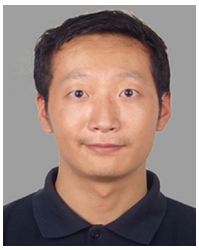
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Hai Yang received the B.Sc. degree in software engineering from Xi'an Jiaotong University, Xi'an, China, in 2008 and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China in 2013.

He is currently a Research Associate Professor with the Department of Computer Science and Engineering, East China University of Science and Technology, Shanghai, China. He has published more than ten articles in renowned international journals, including *Bioinformatics* and *Nature Neuroscience*. His research interests include artificial intelligence, machine learning, big data, and bioinformatics.



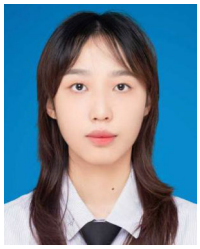
Xingdu Wu is currently pursuing the master's degree in computer technology with East China University of Science and Technology, Shanghai, China.

His research interests include infrared small target detection and multispectral object detection.



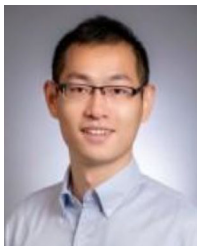
Chenhai Wei is currently pursuing the master's degree in computer technology with East China University of Science and Technology, Shanghai, China.

His research interests include cancer subtyping.



Yunfei Tong is currently pursuing the Ph.D. degree in computer science with East China University of Science and Technology, Shanghai, China.

Her research interests include small target detection and infrared small target tracking.



Dan Zhou received the Ph.D. degree from Zhejiang University School of Medicine, Zhejiang University, Hangzhou, China, in 2017.

He is currently a Professor with The Second Affiliated Hospital of Zhejiang University School of Medicine, Zhejiang University. At present, he has more than 15 papers with the first or corresponding author published on some famous international journals, including *Nature Genetics*, *Nature Communications*, and *Cancer Research*. His research interests include genomics, epidemiology, statistics, bioinformatics, and other disciplines.



Tianyi Zhou received the B.Eng. degree from East China University of Science and Technology, Shanghai, China. He is currently pursuing the master's degree in electrical and computer engineering with the University of Michigan, Ann Arbor, MI, USA, in 2024.

He enjoys building practical AI systems and working across software engineering, data pipelines, and model-driven applications. Recently, he has been exploring large language models and how they can be combined with agent systems to create more capable, reliable, and general-purpose AI tools.



Ting Xiao received the master's and Ph.D. degrees from the School of Computer Science and Technology, Harbin Institute of Technology, Harbin, China, in 2016 and 2022, respectively.

She is currently a Lecturer with East China University of Science and Technology, Shanghai, China. Her research interests include computer vision, transfer learning, few-shot learning, and medical image analysis.

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Qian Zhang received the Ph.D. degree from the School of Computer Science, Beijing Institute of Technology, Beijing, China, in 2019.

She was a Post-Doctoral Researcher with the School of Software, Tsinghua University, Beijing. She is currently a Lecturer with the School of Information Science and Engineering, East China University of Science and Technology, Shanghai, China. Her research interests include mobile computing, smart sensing, and artificial intelligence.



Qin Zhou received the Ph.D. degree from the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2019.

During the Ph.D. degree, she was a Visiting Student with the Professor Haibin Ling's Laboratory, Temple University, Philadelphia, PA, USA. In 2019, she joined Alibaba DAMO Academy, Hangzhou, China as a Senior Algorithm Engineer. She is an Associate Professor with East China University of Science and Technology, Shanghai, China. Previously, she was a Post-Doctoral Researcher with the Robotics Institute, Shanghai Jiao Tong University. Her research interests include multimodal segmentation, partial supervision, few-shot learning, and domain generalization.



Zhe Liu received the B.Sc. degree in computer science from Northeast Normal University, Changchun, China, in 2020, and the Ph.D. degree in biomedical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2025.

She is currently a Lecturer with the Department of Computer Science and Engineering, East China University of Science and Technology, Shanghai. Her research interests include AI for science, focusing on developing interpretable and biologically meaningful intelligent computational models. Specific research topics include variant effect prediction, protein structure and interaction modeling, and the application of artificial intelligence in psychiatric diagnosis and treatment.



Zhe Wang (Associate Member, IEEE) received the B.Sc. and Ph.D. degrees from the Department of Computer Science and Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2003 and 2008, respectively.

He is currently a Full Professor with the Department of Computer Science and Engineering, East China University of Science and Technology, Shanghai, China. At present, he has more than 60 articles with the first or corresponding author published on some famous international journals, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Pattern Recognition*. His research interests include feature extraction, kernel-based methods, image processing, and pattern recognition.

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